

The empirical recurrence for OEIS A268027, recovered from its tail

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8 September 2026

Abstract

OEIS A268027 records “Empirical recurrence of order 86” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 1537$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 88$. Its last coefficient is $c_{86} = 38416 \neq 0$, so no further tail satisfies anything shorter; any order-86 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A268027 is “Number of $3 \times n$ 0..2 arrays with every repeated value in every row greater than or equal to, and in every column greater than, the previous repeated value.”. It has offset 1 and begins

24, 576, 13824, 295962, 6042732, 117582228, 2205234900, 40088968398, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Jan 24 2016 14:11:22, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 1537$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 1537$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 86: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 86.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 3074$ exact terms from $n = 2$ on, it returns order 86 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	51	c_{23}	−311625255499827152014	c_{45}	70082058633584921019074913	c_{67}	7
c_2	−504	c_{24}	−211807988685526882356	c_{46}	11563060542704475052038527	c_{68}	2
c_3	−13928	c_{25}	2527749988245959267547	c_{47}	−69644335190659608451729546	c_{69}	−
c_4	251433	c_{26}	2459284448408541837489	c_{48}	−1507477128164780845777260	c_{70}	−
c_5	1736192	c_{27}	−16357423347178396214794	c_{49}	56171525003582360728370161	c_{71}	−
c_6	−47851091	c_{28}	−18701087425660345869364	c_{50}	−6061649020163979936383325	c_{72}	−
c_7	−149971760	c_{29}	86109687885667617904567	c_{51}	−36409538008355687014642038	c_{73}	−
c_8	5478657085	c_{30}	104698961801367622408937	c_{52}	8250411264739511749662472	c_{74}	−
c_9	13104572687	c_{31}	−373991580824656134176678	c_{53}	18746142585705457623545113	c_{75}	−
c_{10}	−420705092494	c_{32}	−451387125111444867871446	c_{54}	−6362668273936610804278325	c_{76}	−
c_{11}	−1185499003554	c_{33}	1352752982517962408253357	c_{55}	−7557603557683278284058154	c_{77}	−
c_{12}	21926099816252	c_{34}	1530373710064782385360901	c_{56}	3433400728183171410133100	c_{78}	−
c_{13}	82927899759203	c_{35}	−4096181253002188021249418	c_{57}	2341710386319097278157033	c_{79}	−
c_{14}	−747522673981003	c_{36}	−4117759629909978000317398	c_{58}	−1369979693040786373869213	c_{80}	−
c_{15}	−3801763864791838	c_{37}	10399860789716586180075295	c_{59}	−542794140993522448764910	c_{81}	−
c_{16}	15737617954518748	c_{38}	8801675103186475196438327	c_{60}	412434063167013344488484	c_{82}	−
c_{17}	110189996267821411	c_{39}	−22108981125839282709074918	c_{61}	89762099645279524618429	c_{83}	−
c_{18}	−196592219475352399	c_{40}	−14838239101888663094969922	c_{62}	−94477202765280876513661	c_{84}	−
c_{19}	−2145045600569388306	c_{41}	39213957500871750799679805	c_{63}	−9416386394480422307554	c_{85}	−
c_{20}	1130574502086151120	c_{42}	19350242092494324218595753	c_{64}	16535411157325863814191	c_{86}	−
c_{21}	29853326097440969523	c_{43}	−57727311013917088100635142	c_{65}	323502304109928337980		
c_{22}	6407263964876807889	c_{44}	−18640743152713763633896572	c_{66}	−2216147551966236321329		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 88$, and it is the recurrence OEIS A268027 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{86} - c_1x^{85} - \dots - c_{86}$. Its constant term is $-c_{86} = -38416$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 86 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 86, so $\deg q = 86$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 86, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 86, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 38416.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-86 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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